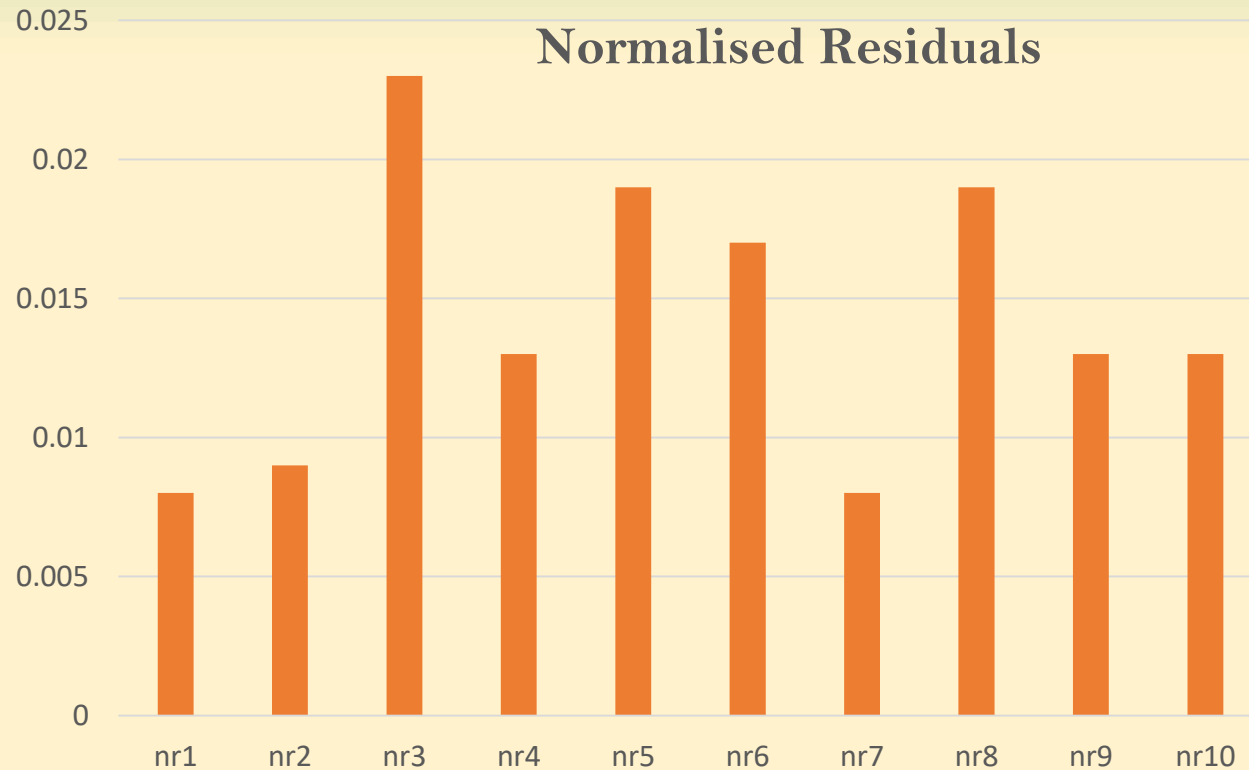
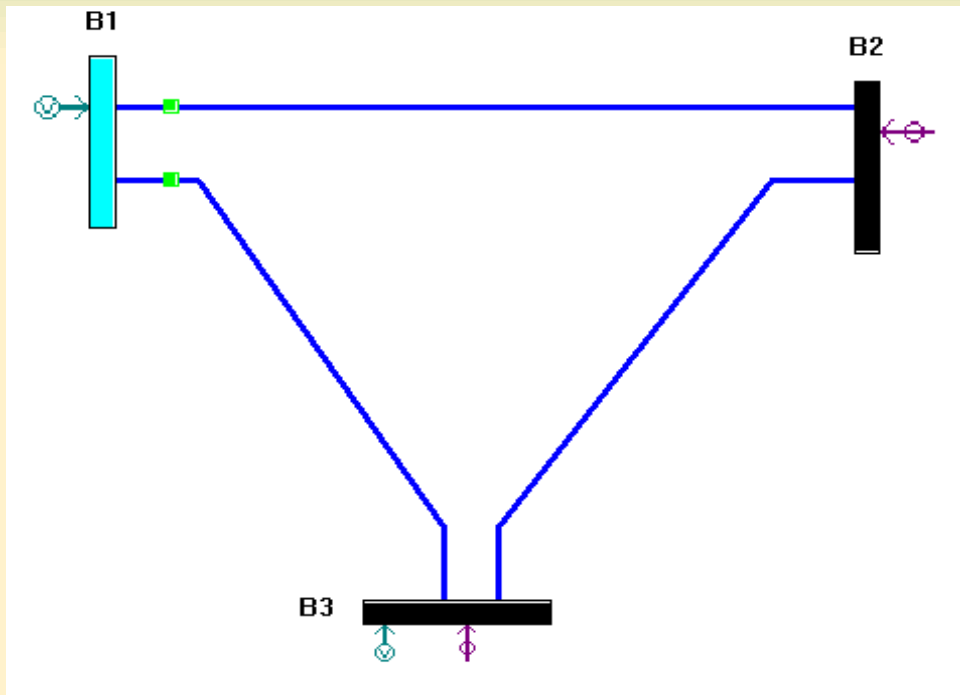


Realistic State Estimator For Energy Management System



Test Cases (3, 14 and 30 bus)

SCADA only

SCADA +
PMU +
Corrupted

Skipped /
Insufficient
Meas.

Supported Measurements

Voltage

- Magnitude (Block 1)
- Angle (Block 2)

Power Injection

- Active (Block 3)
- Reactive (Block 4)

Power Flow

- Active (Block 5)
- Reactive (Block 6)

Current

- Magnitude (Block 7)
- Angle (Block 8)

Errors can be correlated if more than one residuals are large (error is spread)

One Bad data can effect other residuals, it can be removed if the data is redundant

Learning Outcome

Removing Bad data (if it is not critical measurement) reduces almost all residuals

System is unobservable if G is singular or Rank deficient (G or H)

Resources

PET

PyCharm

Claude